Week 8: Database Relationship Analysis Using Joins

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SCENARIO

The Perfume shop maintains information about perfumes, suppliers, customers, sales, and stock details in a database. The management needs an efficient system to store and retrieve information, monitor perfume availability, track sales transactions, and identify perfumes that need to be restocked. SQL queries are used to manage the database, retrieve required records, analyze sales details, and support the smooth and effective daily operations of the Perfume shop.

1. Combine Related Tables Using Joins

- **Concept:** A JOIN is used to retrieve data from two or more related tables using common attributes.

Tables to be Combined

- **Customer Table** *(Stores customer information)*
    - Customer_ID
    - Customer_Name
    - Email
    - City
- **Orders Table** *(Stores customer order details)*
    - Order_ID
    - Customer_ID
    - Order_Date
    - Total_Amount
    - Order_Status
- **Product Table** *(Stores product details)*
    - Product_ID
    - Product_Name
    - Price
    - Category_ID
- **Payment Table** *(Stores payment transactions)*
    - Payment_ID
    - Order_ID
    - Payment_Mode
    - Payment_Status

Implement INNER JOIN

INNER JOIN returns only matching records from both tables.

1. Display customer details along with their orders.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, C.EMAIL, C.CITY, O.* FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                         ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
---------------------------- ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1001     Chiru                Raju
rajuch06@gmail.com
Chennai                      ORD001  C-1001  25-SEP-26         2500
Delivered


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                         ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
---------------------------- ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1002     Hema                 Bhaskar
hemabhaskar02@gmail.com
Coimbatore                   ORD002  C-1002  25-SEP-26         1800
Pending


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                         ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
---------------------------- ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1003     Seetha               Mani
seethamani16@gmail.com
Madurai                      ORD003  C-1003  24-SEP-26         3500
Shipped
```

```
CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                         ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
---------------------------- ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1003     Seetha               Mani
seethamani16@gmail.com
Madurai                      ORD003  C-1003  24-SEP-26         3500
Shipped


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                         ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
---------------------------- ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1004     Abinaya              Ramesh
abiramesh79@gmail.com
Salem                        ORD004  C-1004  23-SEP-26         1200
Delivered
```

2. Display order details with payment information.

```
SQL> SELECT O.*, P.PAYMENT_ID, P.PAYMENT_MODE, P.PAYMENT_STATUS FROM ORDERS O INNER JOIN PAYMENT P ON O.ORDER_ID = P.ORDER_ID;

ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
------- ------- --------- ------------ -------------------- ----------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
ORD001  C-1001  25-SEP-26         2500 Delivered            PAY001
UPI                  Success

ORD002  C-1002  25-SEP-26         1800 Pending              PAY002
Card                 Pending

ORD003  C-1003  24-SEP-26         3500 Shipped              PAY003
Net Banking          Success


ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
------- ------- --------- ------------ -------------------- ----------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
ORD004  C-1004  23-SEP-26         1200 Delivered            PAY004
Cash                 Success

ORD005  C-1005  22-SEP-26         4500 Pending              PAY005
UPI                  Failed
```

3. Retrieve products purchased by customers.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, C.EMAIL FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.CUSTOMER_ID IS NULL;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
C-1006     Sanjana              Doss
sanjanat08@gmail.com

C-1007     Vanaja               Muniyappan
vanajam2008@gmail.com

C-1008     Priya                Hari
haripriya2007@gmail.com


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
C-1009     Bhavana              Karthik
bhavanak29@gmail.com
```

Implement LEFT JOIN

1. Display all customers including customers who have not placed orders.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, C.EMAIL, C.CITY, O.* FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                           ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
------------------------------ ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1001     Chiru                Raju
rajuch06@gmail.com
Chennai                        ORD001  C-1001  25-SEP-26         2500
Delivered


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                           ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
------------------------------ ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1002     Hema                 Bhaskar
hemabhaskar02@gmail.com
Coimbatore                     ORD002  C-1002  25-SEP-26         1800
Pending


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                           ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
------------------------------ ------- ------- --------- ------------
ORDER_STATUS
--------------------
C-1003     Seetha               Mani
seethamani16@gmail.com
Madurai                        ORD003  C-1003  24-SEP-26         3500
Shipped
```

2. Find customers without any purchases.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, C.EMAIL FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.CUSTOMER_ID IS NULL;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
C-1006     Sanjana              Doss
sanjanat08@gmail.com

C-1007     Vanaja               Muniyappan
vanajam2008@gmail.com

C-1008     Priya                Hari
haripriya2007@gmail.com


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
C-1009     Bhavana              Karthik
bhavanak29@gmail.com
```

3. Display all products including products with no sales.

```
SQL> SELECT B.*, OD.ORDER_ID FROM BRAND B LEFT JOIN ORDER_DETAILS OD ON B.PRODUCT_ID = OD.PRODUCT_ID;

PRODUCT_ID BRAND_NAME                    CATEGORY_I STOCK_QUANTITY      PRICE
---------- ------------------------------ ---------- -------------- ----------
PRODUCT_COUNT ORDER_I
------------- -------
BR-01      Dior                          CAT-001                25       1500
            1 ORD001

BR-02      Chanel                        CAT-002                30       1800
            1 ORD002

BR-03      Gucci                         CAT-003                20       1200
            1 ORD003


PRODUCT_ID BRAND_NAME                    CATEGORY_I STOCK_QUANTITY      PRICE
---------- ------------------------------ ---------- -------------- ----------
PRODUCT_COUNT ORDER_I
------------- -------
BR-04      Versace                       CAT-004                15       2200
            1 ORD004

BR-05      Armani                        CAT-005                35       1400
            1 ORD005

BR-10      Fogg                          CAT-010                24       2000
            1


PRODUCT_ID BRAND_NAME                    CATEGORY_I STOCK_QUANTITY      PRICE
---------- ------------------------------ ---------- -------------- ----------
PRODUCT_COUNT ORDER_I
------------- -------
BR-07      Calvin                        CAT-007                10       3500
            1

BR-06      Burberry                      CAT-006                28       1600
            1

BR-08      Hugo Boss                     CAT-008                22       1900
            1


PRODUCT_ID BRAND_NAME                    CATEGORY_I STOCK_QUANTITY      PRICE
---------- ------------------------------ ---------- -------------- ----------
PRODUCT_COUNT ORDER_I
------------- -------
BR-09      YSL                           CAT-009                18       2500
            1


10 rows selected.
```

Implement RIGHT JOIN

1. Display all orders with customer information about their products.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, C.EMAIL, O.* FROM CUSTOMER C RIGHT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                              ORDER_I CUSTOME ORDER_DAT
-------------------------------------------------- ------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
C-1001     Chiru                Raju
rajuch06@gmail.com                                 ORD001  C-1001  25-SEP-26
        2500 Delivered

C-1002     Hema                 Bhaskar
hemabhaskar02@gmail.com                            ORD002  C-1002  25-SEP-26
        1800 Pending

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                              ORDER_I CUSTOME ORDER_DAT
-------------------------------------------------- ------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------

C-1003     Seetha               Mani
seethamani16@gmail.com                             ORD003  C-1003  24-SEP-26
        3500 Shipped

C-1004     Abinaya              Ramesh
abiramesh79@gmail.com                              ORD004  C-1004  23-SEP-26

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                              ORDER_I CUSTOME ORDER_DAT
-------------------------------------------------- ------- ------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
        1200 Delivered

C-1005     Deepikaa             Selvam
deepiselva30@gmail.com                             ORD005  C-1005  22-SEP-26
        4500 Pending
```

2. Find orders where customer details are missing.

```
SQL> SELECT O.* FROM CUSTOMER C RIGHT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID IS NULL;

no rows selected
```

3. Display all payment records with order details.

```
SQL> SELECT O.* FROM CUSTOMER C RIGHT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID IS NULL;

no rows selected

SQL> SELECT P.*, O.* FROM ORDERS O RIGHT JOIN PAYMENT P ON O.ORDER_ID = P.ORDER_ID;

PAYMENT_ID ORDER_I PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ------- --------- -------------------- --------------------
TRANSACTION_AMOUNT ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------------ ------- ------- --------- ------------ --------------------
PAY001     ORD001  25-SEP-26 UPI                 Success
              2500 ORD001  C-1001  25-SEP-26         2500 Delivered

PAY002     ORD002  25-SEP-26 Card                Pending
              1800 ORD002  C-1002  25-SEP-26         1800 Pending

PAY003     ORD003  24-SEP-26 Net Banking         Success
              3500 ORD003  C-1003  24-SEP-26         3500 Shipped


PAYMENT_ID ORDER_I PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ------- --------- -------------------- --------------------
TRANSACTION_AMOUNT ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------------ ------- ------- --------- ------------ --------------------
PAY004     ORD004  23-SEP-26 Cash                Success
              1200 ORD004  C-1004  23-SEP-26         1200 Delivered

PAY005     ORD005  22-SEP-26 UPI                 Failed
              4500 ORD005  C-1005  22-SEP-26         4500 Pending
```

5. Retrieve Complete Order Details Task:

Generate a complete order report by combining:

Customer Name

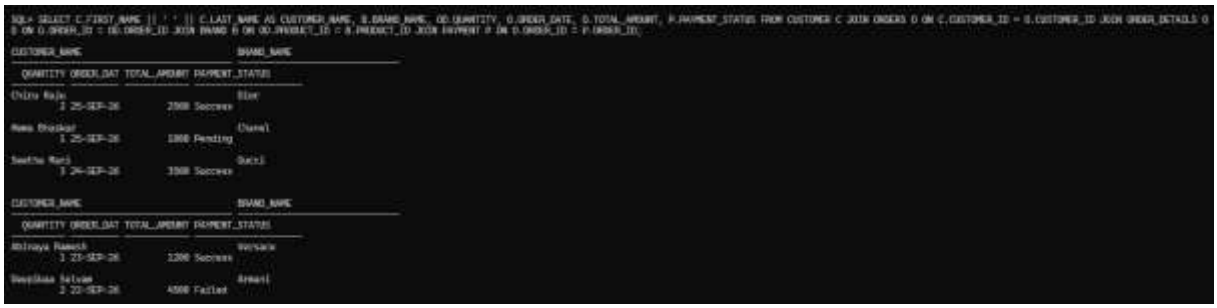
Product Name

Quantity

Order Date

Total Amount

Payment Status



6. Display Customer Purchase History

1. All products purchased by a customer.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, B.BRAND_NAME FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_DETAILS
  OD ON O.ORDER_ID = OD.ORDER_ID JOIN BRAND B ON OD.PRODUCT_ID = B.PRODUCT_ID WHERE C.CUSTOMER_ID = '&CUSTOMER_ID';
Enter value for customer_id: C-1003
old   1: SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, B.BRAND_NAME FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_DET
AILS OD ON O.ORDER_ID = OD.ORDER_ID JOIN BRAND B ON OD.PRODUCT_ID = B.PRODUCT_ID WHERE C.CUSTOMER_ID = '&CUSTOMER_ID'
new   1: SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, B.BRAND_NAME FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_DET
AILS OD ON O.ORDER_ID = OD.ORDER_ID JOIN BRAND B ON OD.PRODUCT_ID = B.PRODUCT_ID WHERE C.CUSTOMER_ID = 'C-1003'

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
BRAND_NAME
------------------------------
C-1003     Seetha               Mani
Gucci
```

2. Total amount spent by each customer.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER
_ID GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME;

CUSTOMER_I FIRST_NAME           LAST_NAME            TOTAL_SPENT
---------- -------------------- -------------------- -----------
C-1001     Chiru                Raju                        2500
C-1002     Hema                 Bhaskar                     1800
C-1005     Deepikaa             Selvam                      4500
C-1003     Seetha               Mani                        3500
C-1004     Abinaya              Ramesh                      1200
```

3. Number of orders placed by each customer

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, COUNT(O.ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUST
OMER_ID GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME;

CUSTOMER_I FIRST_NAME           LAST_NAME            TOTAL_ORDERS
---------- -------------------- -------------------- ------------
C-1008     Priya                Hari                            0
C-1009     Bhavana              Karthik                         0
C-1001     Chiru                Raju                            1
C-1002     Hema                 Bhaskar                         1
C-1007     Vanaja               Muniyappan                      0
C-1005     Deepikaa             Selvam                          1
C-1006     Sanjana              Dass                            0
C-1003     Seetha               Mani                            1
C-1004     Abinaya              Ramesh                          1

9 rows selected.
```

```
SQL> SELECT * FROM (SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C.CUS
TOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID = '&CUSTOMER_ID' ORDER BY O.ORDER_DATE DESC) WHERE ROWNUM = 1;
Enter value for customer_id: C-1002
old   1: SELECT * FROM (SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C
.CUSTOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID = '&CUSTOMER_ID' ORDER BY O.ORDER_DATE DESC) WHERE ROWNUM = 1
new   1: SELECT * FROM (SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C
.CUSTOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID = 'C-1002' ORDER BY O.ORDER_DATE DESC) WHERE ROWNUM = 1

CUSTOMER_I FIRST_NAME           LAST_NAME            ORDER_I ORDER_DAT
---------- -------------------- -------------------- ------- ---------
TOTAL_AMOUNT
------------
C-1002     Hema                 Bhaskar              ORD002  25-SEP-26
      1800
```

4. Latest purchase details of customers.

```
SQL> SELECT * FROM (SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C.CUS
TOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID = '&CUSTOMER_ID' ORDER BY O.ORDER_DATE DESC) WHERE ROWNUM = 1;
Enter value for customer_id: C-1002
old   1: SELECT * FROM (SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C
.CUSTOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID = '&CUSTOMER_ID' ORDER BY O.ORDER_DATE DESC) WHERE ROWNUM = 1
new   1: SELECT * FROM (SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C
.CUSTOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOMER_ID = 'C-1002' ORDER BY O.ORDER_DATE DESC) WHERE ROWNUM = 1

CUSTOMER_I FIRST_NAME           LAST_NAME            ORDER_I ORDER_DAT
---------- -------------------- -------------------- ------- ---------
TOTAL_AMOUNT
------------
C-1002     Hema                 Bhaskar              ORD002  25-SEP-26
      1800
```

7. Generate Multi-Table Business Report

Report 1: Customer Order Report

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, O.ORDER_ID, O.ORDER_DATE, O.ORDER_STATUS FROM CUSTOMER C INNER JOIN ORDERS O ON C.C
USTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME                          ORDER_I ORDER_DAT ORDER_STATUS
-------------------------------------- ------- --------- --------------------
Chiru Raju                             ORD001  25-SEP-26 Delivered
Hema Bhaskar                           ORD002  25-SEP-26 Pending
Seetha Mani                            ORD003  24-SEP-26 Shipped
Abinaya Ramesh                         ORD004  23-SEP-26 Delivered
Deepikaa Selvam                        ORD005  22-SEP-26 Pending
```

Report 2: Sales Report

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, O.ORDER_ID, O.ORDER_DATE, O.ORDER_STATUS FROM CUSTOMER C INNER JOIN ORDERS O ON C.C
USTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME                          ORDER_I ORDER_DAT ORDER_STATUS
-------------------------------------- ------- --------- --------------------
Chiru Raju                             ORD001  25-SEP-26 Delivered
Hema Bhaskar                           ORD002  25-SEP-26 Pending
Seetha Mani                            ORD003  24-SEP-26 Shipped
Abinaya Ramesh                         ORD004  23-SEP-26 Delivered
Deepikaa Selvam                        ORD005  22-SEP-26 Pending
```

Report 3: Payment Analysis Report

```
SQL> SELECT B.BRAND_NAME, SUM(OD.QUANTITY) AS QUANTITY_SOLD, SUM(OD.QUANTITY * B.PRICE) AS TOTAL_REVENUE FROM BRAND B INNER JOIN ORDER_DETAILS OD ON
 B.PRODUCT_ID = OD.PRODUCT_ID GROUP BY B.BRAND_NAME;

BRAND_NAME                     QUANTITY_SOLD TOTAL_REVENUE
------------------------------ ------------- -------------
Chanel                                     1          1800
Armani                                     2          2800
Dior                                       2          3800
Gucci                                      3          3600
Versace                                    1          2200
```

Report 4: Customer Purchase Analysis

(top purchasing order)

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.
CUSTOMER_ID, C.CUSTOMER_NAME ORDER BY TOTAL_SPENDING DESC;

CUSTO CUSTOMER_NAME                    TOTAL_SPENDING
----- ------------------------------ ---------------
C001  Devi                                     5000
C007  Sindhu                                   4500
C002  Radhi                                    4400
C004  Vijay                                    3200
C005  Abi                                      2750
C009  Sanjay                                   2100
C003  Ajay                                     1800
C006  Anu                                      1200
```

Maximum orders

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, COUNT(O.ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUST
OMER_ID, C.CUSTOMER_NAME HAVING COUNT(O.ORDER_ID) = (SELECT MAX(ORDER_COUNT) FROM (SELECT COUNT(ORDER_ID) AS ORDER_COUNT FROM ORDERS GROUP BY CUSTOMER_ID));

CUSTO CUSTOMER_NAME                    TOTAL_ORDERS
----- ------------------------------ ------------
C001  Devi                                      2
C002  Radhi                                     2
```

Customers with highest spending

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, ROUND(SUM(O.TOTAL_AMOUNT), 2) AS TOTAL_AMOUNT_SPENT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_
ID GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME HAVING SUM(O.TOTAL_AMOUNT) = (SELECT MAX(TOTAL_SPENT) FROM (SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENT FROM ORDERS G
ROUP BY CUSTOMER_ID));

CUSTO CUSTOMER_NAME                    TOTAL_AMOUNT_SPENT
----- ------------------------------ ------------------
C001  Devi                                         5000
```